WaferEngine · CS-3

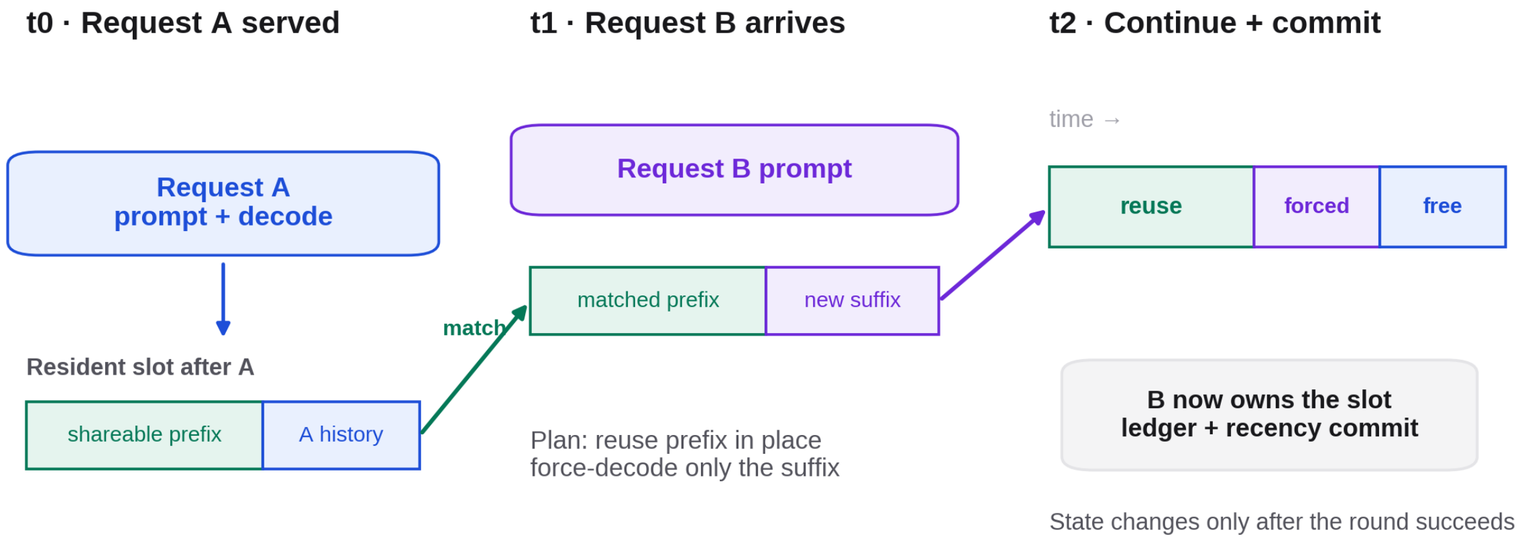
WEEKLY COLLABORATOR DISCUSSION

In-PE KV Reuse

Measured benefits, execution constraints, and an on-wafer secondary KV tier.

MODEL	EVIDENCE	SCOPE	DATE
Qwen3-1.7B	Real CS-3 TSC	M1 reuse + M3 exploration	2026-08-11

A later request can reuse KV left by an earlier request



Example: B shares A's resident system prefix; the appliance skips that work, rebuilds only B's suffix, then decodes normally.

One slot across three time steps: match the resident prefix, force only the new suffix, then continue normal decode and commit the new resident history.

CURRENT FEATURE SURFACE

Resident KV is now a serving resource, not a one-request scratchpad

ROUND BEHAVIOR

Reuse, rebuild, or replace

The host chooses a one-to-one resident slot assignment for the batch. The appliance starts from a shared safe prefix, receives no host KV payload on this path, and extends the slot with real prompt and sampled tokens.

ENABLED SCENARIOS

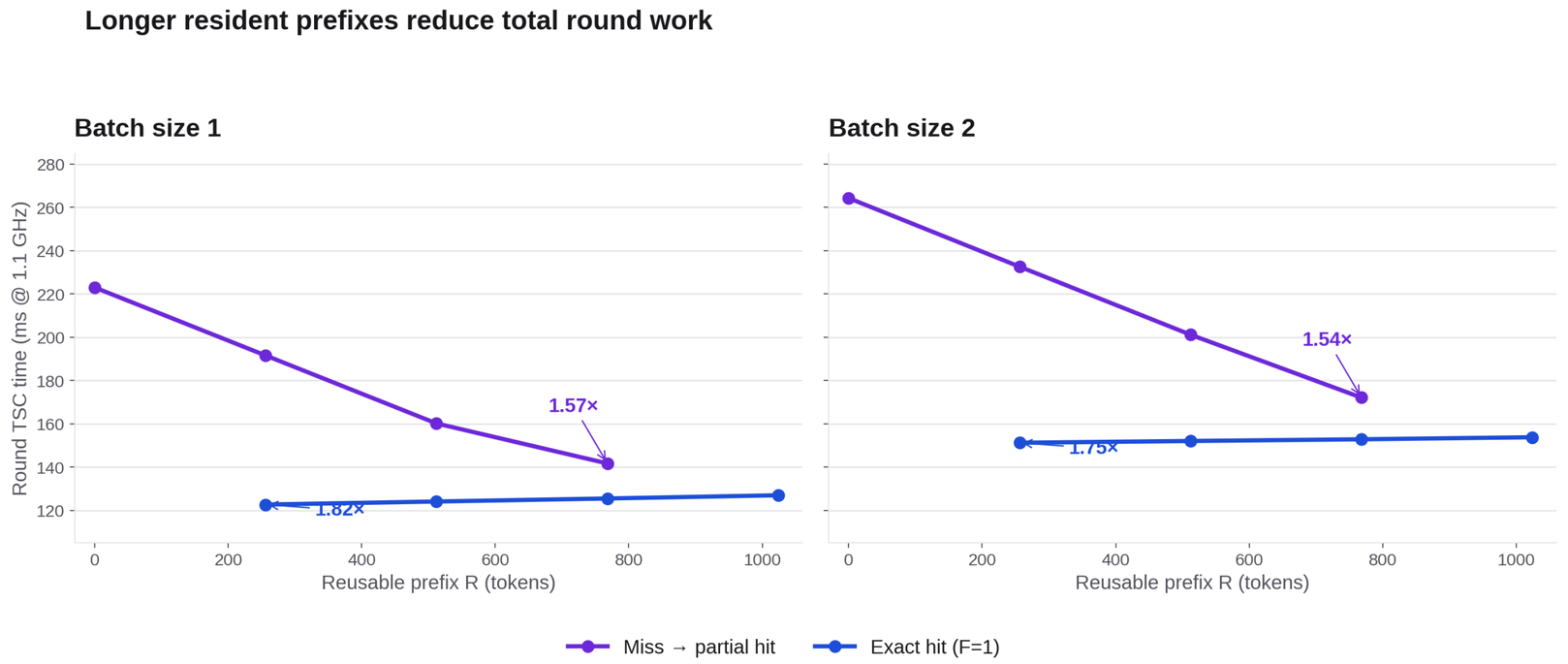
01 Hit — reuse in place; exact retains one seed, prefix forces only prompt[start:]

02 Miss — start at zero and rebuild the prompt by force-decode

03 Pressure — use an empty slot, otherwise replace one whole slot by LRU

EXPERIMENT A · REUSE LENGTH × BATCH

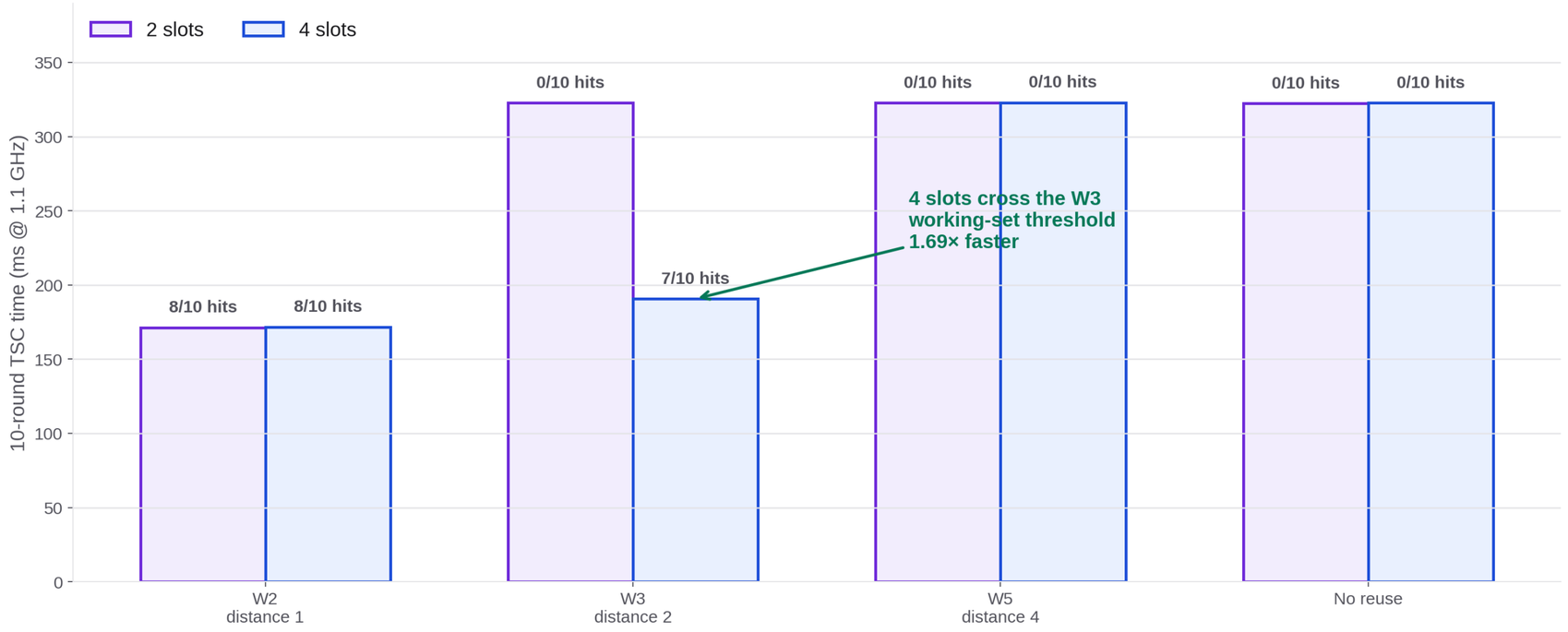
More reusable prefix translates into less round work



Full Qwen3-1.7B, real WSE-3, three repetitions per point. bsz 3–4 reached the PE-memory compile boundary (no TSC); partial shortens the suffix, while exact holds F=1.

EXPERIMENT B · SYNTHETIC LOCALITY × SLOTS

More slots help only when they cross the working-set threshold



Ten-round synthetic traces, fresh sequence IDs, 256-token shared prefixes. W3 thrashes with two slots but produces seven content hits with four slots: 1.69× lower TSC than no reuse.

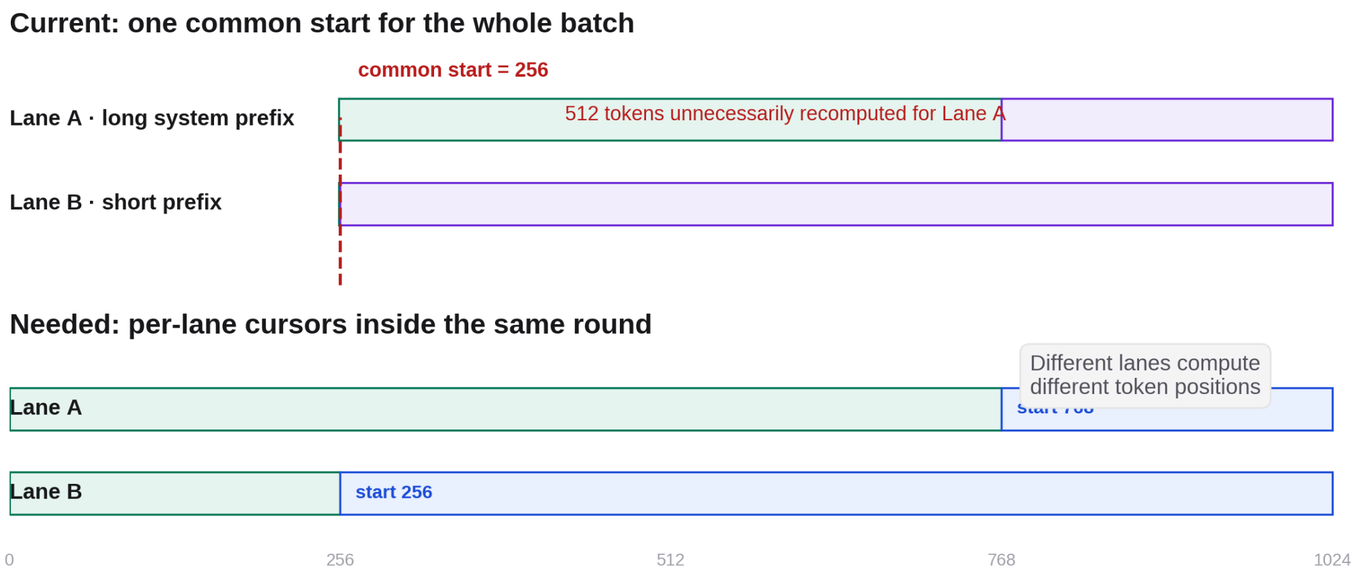
LIMITATION

Matching works; divergent execution is still missing.

Today every lane in a round is forced onto one common safe start, even when its reusable prefix is longer.

LIMITATION A · RAGGED EXECUTION

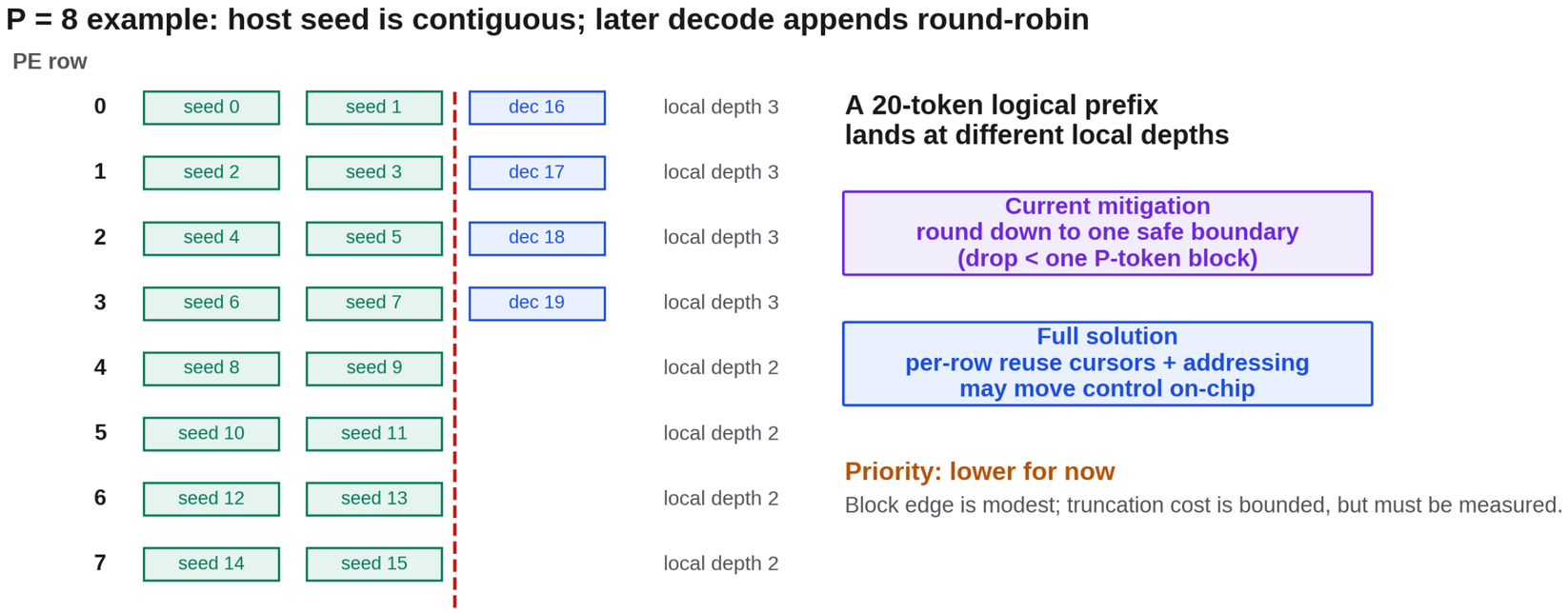
One shared start is unfair to requests with long reusable prefixes



Example: Lane A can reuse 768 tokens and Lane B only 256. The current batch starts both at 256; ragged execution would let the same PE block compute different token positions per lane.

LIMITATION B · PER-ROW KV LAYOUT

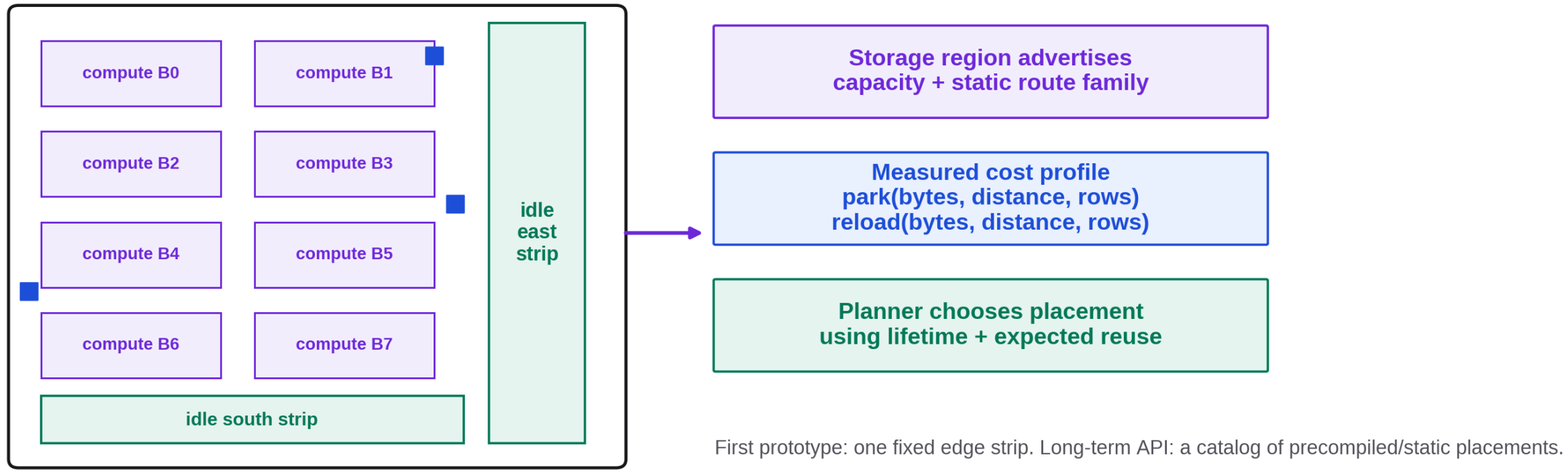
Seed and decode KV can produce different local reuse depths



After a contiguous host seed and round-robin decode appends, one logical prefix can end at different local indices across PE rows. The current safe mitigation truncates to a shared boundary.

Support flexible placement through measured, static movement profiles

Direction: placement-agnostic policy, statically realizable movement



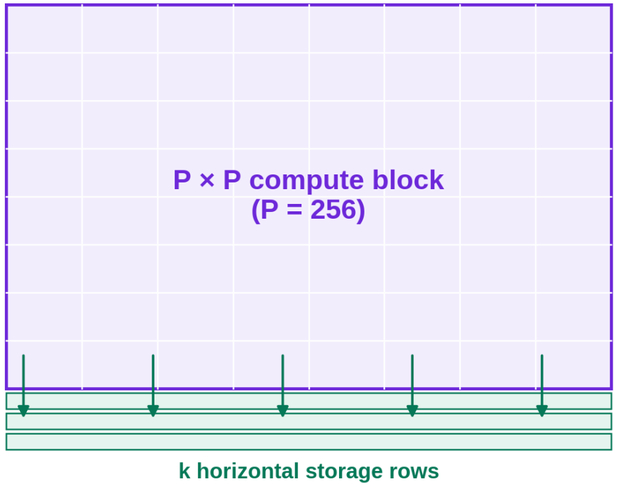
First prototype: one fixed edge strip. Long-term API: a catalog of precompiled/static placements.

Candidate storage can live in any usable region — but routes are not keyed at runtime.

Logical placement can be general, but WSE routes are not keyed by request ID. Each usable storage region should expose capacity plus measured park/reload curves for a precompiled route family.

A horizontal storage strip gives a simple first mechanism—and a clear capacity bill

Concrete probe: attach horizontal storage rows under each PE block



1 row × 256 PEs × 42 KiB
≈ 10.5 MiB per block

Full-model KV density per block
≈ 16 KiB / token (8 blocks)

One row stores ≈ 672 tokens
≈ 13 rows for one 8K request

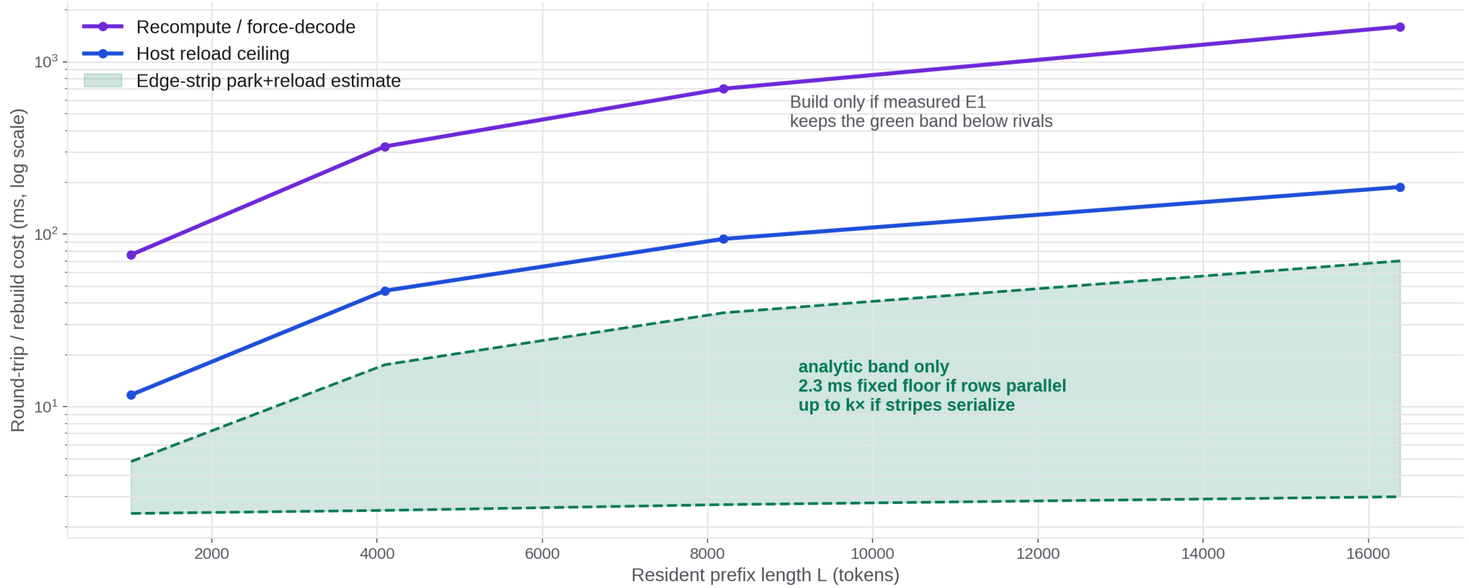
All numbers are analytical and use the nominal 42 KiB storage payload; E2 must compile-probe the real cap.

Open: can k rows move concurrently without stealing live colors/queues from decode?

Nominal analytical sizing: 42 KiB payload per storage PE and 128 KiB/token across the eight-block model. One 1×256 strip stores about 672 tokens per block; an 8K request needs about 13 rows.

M3 COST MODEL · DISCUSSION, NOT A MEASUREMENT

Eviction is attractive only if the movement mechanism lands in the green band



Analytical edge-strip range: 2×P×4.54 µs ≈ 2.3 ms fixed round-trip if storage rows move in parallel; up to k× if rows serialize. E1 must measure payload, distance, row scaling and both directions.

QUESTIONS FOR COLLABORATORS

What should we resolve next?

Ragged	Prioritize per-lane starts; choose the long-prefix success workload.
Row layout	Measure bounded reuse loss before moving per-row addressing on-chip.
M3-E1	Sweep park/reload by payload, distance, active rows and direction.
Placement + policy	Prototype one profiled edge strip; let measured cost drive eviction.